

Azure data Factory

- **What is ADF?**

Azure Data Factory (ADF) is a cloud-based data integration service from Microsoft that allows you to create, schedule, and orchestrate data workflows at scale. It is designed for Extract, Transform, Load (ETL) and Extract, Load, Transform (ELT) processes, enabling seamless movement and transformation of data across various sources and destinations.

Main Features of Azure Data Factory:

1. Data Ingestion & Integration

- Supports over 90 built-in connectors (e.g., SQL Server, Azure Blob Storage, Amazon S3, SAP, Salesforce, etc.).
- Enables hybrid data integration, connecting both on-premises and cloud data sources.

2. Data Transformation (ETL/ELT)

- Provides **Mapping Data Flows** for code-free transformations.
- Supports **Azure Databricks, Azure Synapse Analytics, HDInsight, and Data Lake** for advanced data transformations.
- Can leverage custom processing using **Azure Functions, Logic Apps, or Machine Learning models**.

3. Pipeline Orchestration & Scheduling

- Automates data workflows using **pipelines** with dependencies and triggers.
- Supports event-driven execution (e.g., based on file arrival or database changes).

4. Data Flow Monitoring & Logging

- Built-in monitoring with **Azure Monitor and Application Insights** integration.
- Provides real-time and historical run-time analytics for debugging and optimization.

5. Scalability & Performance

- Serverless architecture scales automatically based on workload demands.
- Optimized for high-performance big data processing.

6. Security & Compliance

- Supports **Azure Key Vault** for secrets management.
- Provides role-based access control (RBAC), Virtual Network (VNet) integration, and encryption for secure data handling.

7. Cost-Effectiveness

- Pay-as-you-go pricing model with metered usage.
- Reduces infrastructure management overhead by using managed services.

Benefits of Azure Data Factory:

- **Simplifies data movement and transformation** without requiring extensive coding.
- **Enhances data-driven decision-making** by unifying data from multiple sources.
- **Integrates seamlessly** with other Azure services like Azure Synapse, Azure ML, and Power BI.
- **Optimizes operational efficiency** through automation and scheduled workflows.
- **Ensures enterprise-grade security** and compliance for data governance.

What are the main components of Azure Data Factory?

Azure Data Factory (ADF) consists of several key components that work together to enable data integration, transformation, and orchestration. Here are the **main components** of ADF:

1. Pipelines

A **pipeline** is a logical grouping of activities that perform a specific data workflow.

- It allows you to **ingest, transform, and move data** from various sources to destinations.
- You can have **multiple activities** in a single pipeline, such as data copy, transformations, and control flow operations.

Example: A pipeline that copies data from an on-premises SQL database to Azure Data Lake and then transforms it using Azure Databricks.

2. Activities

Activities represent **the tasks** performed within a pipeline. There are three main types:

a. Data Movement Activities

Used to transfer data from source to destination.

Example: Copy Activity moves data from an on-premises SQL Server to Azure Blob Storage.

b. Data Transformation Activities

Used for processing and transforming data.

Example: Data Flow Activity performs transformations using ADF's Mapping Data Flows, or **Databricks Notebook Activity** for complex processing.

c. Control Flow Activities

Used for workflow logic (e.g., looping, conditions, triggers).

Example: ForEach Activity iterates through a list of files and processes them one by one.

3. Datasets

A **dataset** represents the data structure within a source or destination. It defines **what data** will be used in an activity.

Example:

- An **Azure Blob dataset** might point to a folder containing CSV files.
 - An **SQL dataset** might represent a specific table in an Azure SQL Database.
-

4. Linked Services

A **linked service** is a connection to an external data source. It stores connection strings, authentication details, and endpoint configurations.

Example Linked Services:

- **Azure Blob Storage** (for storing files)
 - **SQL Server** (for structured data)
 - **Amazon S3, SAP, Salesforce** (third-party data sources)
-

5. Integration Runtimes (IR)

Integration Runtime (IR) is the compute infrastructure used for **data movement and transformation** in ADF. There are three types:

a. Azure Integration Runtime

- Fully managed and runs in the cloud.

- Used for cloud-to-cloud data transfers and transformations.

b. Self-Hosted Integration Runtime

- Installed on-premises or on a VM.
- Used for **on-premises to cloud** or **hybrid** data movement.
- Required for connecting to **on-prem SQL Server, Oracle, or file shares**.

c. Azure SSIS Integration Runtime

- Used for running **SQL Server Integration Services (SSIS) packages** in the cloud.
-

6. Triggers

Triggers allow **scheduling and automation** of pipelines.

Types of triggers include:

- **Schedule Trigger** – Runs at a predefined time.
- **Event Trigger** – Runs when a file is added to a storage location.
- **Tumbling Window Trigger** – Runs at periodic intervals for batch processing.

Example: A pipeline that runs every morning at 6 AM to extract sales data.

7. Monitoring & Debugging

ADF provides built-in **monitoring tools** to track pipeline executions, performance, and failures.

- **Activity Runs** show step-by-step execution logs.
- **Alerts & Logging** can be set up via **Azure Monitor**.
- **Debug Mode** allows testing pipelines before deployment.

what are different types of triggers and explain?

Azure Data Factory (ADF) provides **triggers** to automate pipeline execution based on time, events, or recurrence patterns. There are **three main types** of triggers in ADF:

1. Schedule Trigger

- Runs a pipeline at a **predefined date and time**.

- Can be set to execute **once** or on a **recurring basis** (e.g., daily, weekly, hourly).
- Useful for **batch processing** or **time-based jobs**.

Example Use Case:

A pipeline that extracts daily sales data from an **SQL database** and loads it into **Azure Data Lake** every night at 12 AM.

Key Properties:

- Start time & End time (optional)
 - Recurrence frequency (e.g., every 1 hour, every day)
 - Time zone support
-

2. Tumbling Window Trigger

- Runs at **fixed time intervals** with a **defined size** (e.g., every 15 minutes, every 2 hours).
- Ensures **no overlap or gaps** in execution.
- Used for **incremental data processing** or **windowed batch processing**.

Example Use Case:

Processing IoT sensor data every **30 minutes** and aggregating it in **Azure Synapse Analytics**.

Key Properties:

- Window size (e.g., 15 min, 1 hour, 1 day)
 - Start time
 - Retry policy (if a window fails, it can be retried)
 - Dependency support (ensures previous windows are completed before the next run)
-

3. Event-Based Trigger: note

- Executes a pipeline **when a specific event occurs** in **Azure Storage**.
- Supports:
 - **Blob storage event triggers** (runs when a file is created or deleted).
 - **Event Grid-based triggers** (listens for custom events in Azure).

Example Use Case:

A pipeline that starts when a **new CSV file** is uploaded to **Azure Blob Storage**, then processes and moves it to an **SQL database**.

Key Properties:

- Event type (e.g., **Blob Created** or **Blob Deleted**)
- Storage account and container monitoring
- Filters (e.g., run only when files match **.csv**)

Comparison Table of ADF Trigger Types:

Trigger Type	When It Runs	Use Case	Key Feature
Schedule Trigger	At a specific time	Daily ETL jobs	Time-based execution
Tumbling Window Trigger	Fixed intervals (non-overlapping)	Batch processing, IoT streaming	Ensures sequential execution
Event-Based Trigger	When an event occurs (file created/deleted)	Real-time data ingestion	Responds to Azure Blob events

Which Trigger Should You Use?

- **Use Schedule Triggers** for fixed, predictable jobs (e.g., daily data loads).
- **Use Tumbling Window Triggers** for batch jobs that process data in chunks (e.g., hourly logs).
- **Use Event-Based Triggers** for real-time processing when new files arrive (e.g., on-demand ETL).

what is integration run time? how many types of integration run times? Explain?

What is Integration Runtime (IR) in Azure Data Factory?

Integration Runtime (IR) is the compute infrastructure used by **Azure Data Factory (ADF)** to execute data movement, transformation, and pipeline activities. It serves as the execution engine that enables connectivity between various data sources, whether cloud-based, on-premises, or hybrid.

Key Functions of Integration Runtime:

- ✓ Moves data between different sources and destinations
 - ✓ Performs data transformations (e.g., Mapping Data Flows)
 - ✓ Supports secure communication with on-premises systems
 - ✓ Executes SQL Server Integration Services (SSIS) packages
-

Types of Integration Runtimes in ADF

Azure Data Factory provides three types of Integration Runtime:

1. Azure Integration Runtime (Azure IR)

- Cloud-based, fully managed runtime provided by Microsoft.
- Used for cloud-to-cloud data movement and transformation.
- Serverless and automatically scales as needed.
- Supports Mapping Data Flows for data transformation.

Use Cases:

- ✓ Moving data between Azure SQL Database and Azure Blob Storage
- ✓ Transforming data using Mapping Data Flows
- ✓ Copying data from Amazon S3 to Azure Data Lake

Key Features:

- No infrastructure management required
 - High availability and auto-scaling
 - Supports copy, data flow, and transformation activities
-

2. Self-Hosted Integration Runtime (SHIR)

- Installed on-premises or on a virtual machine (VM).
- Enables secure hybrid data movement between on-premises and cloud.
- Allows access to on-premises databases, private networks, or file shares.
- Required when connecting to network-isolated data sources (e.g., VNet, VPN, ExpressRoute).

Use Cases:

- ✓ Copying data from on-premises SQL Server to Azure Blob Storage
- ✓ Connecting to an Oracle database behind a firewall
- ✓ Moving files from an internal file server to Azure

Key Features:

- Provides secure data access to on-premises and private network data sources
- Can be scaled out by installing multiple nodes
- Supports copy, lookup, stored procedure execution, and compression
- Data encryption ensures security during movement

3. Azure SSIS Integration Runtime

- Used to lift and shift SQL Server Integration Services (SSIS) workloads to Azure.
- Enables running existing SSIS packages in the cloud without modifications.
- Ideal for organizations that want to migrate on-prem SSIS ETL processes to Azure.

Use Cases:

- ✓ Running SSIS ETL packages in Azure
- ✓ Migrating on-premises SSIS workflows without rewriting them
- ✓ Using custom SSIS components (e.g., script tasks, third-party connectors)

Key Features:

- Fully managed SSIS runtime environment in the cloud
- Supports scaling up/down SSIS package execution
- Compatible with Azure SQL Managed Instance and SQL Server

Comparison of Integration Runtime Types

Integration Runtime	Location	Best For	Key Features
Azure IR	Cloud	Cloud-to-cloud data movement & transformations	Serverless, auto-scaling, fully managed
Self-Hosted IR	On-premises / VM	Hybrid scenarios (on-prem to cloud)	Secure data access, requires manual setup

Integration Runtime	Location	Best For	Key Features
Azure SSIS IR	Cloud	Running SSIS packages in Azure	Fully managed SSIS, supports lift-and-shift migration